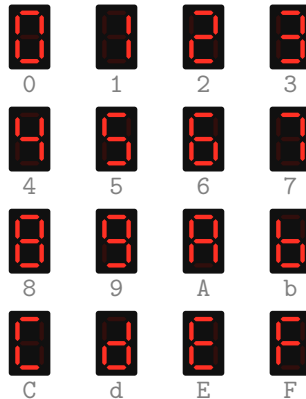
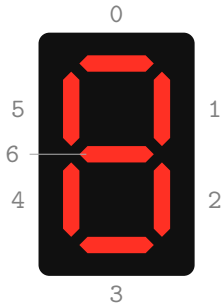


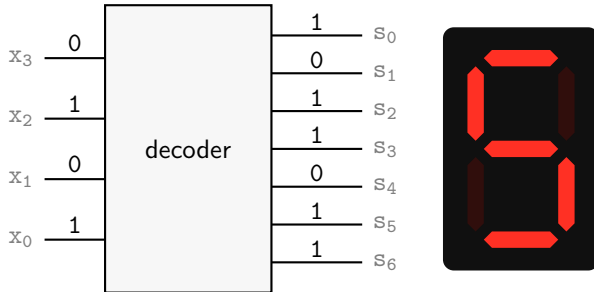
seven-segment decoders

A **seven-segment display** draws a digit by lighting some of seven LEDs, numbered 0 to 6. Different patterns give all sixteen hex digits.



b and d are lower case, so they aren't mistaken for 8 and 0

A **seven-segment decoder** reads a 4-bit number and outputs the seven segment signals, s_0 to s_6 , that draw it.



input 0101 lights segments 0, 2, 3, 5 and 6, drawing a 5

Each segment is its own function of the four inputs. Segment 0 is lit for every digit except 1, 4, b and d.

segment 0

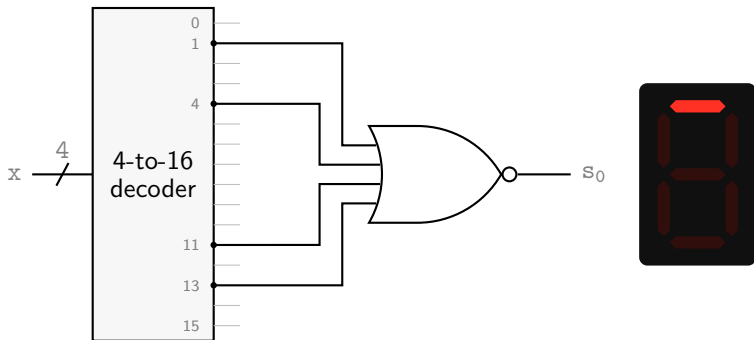


x		S ₀
0000		1
0001		0
0010		1
0011		1
0100		0
0101		1
0110		1
0111		1

x		S ₀
1000		1
1001		1
1010		1
1011		0
1100		1
1101		0
1110		1
1111		1

the other six segments each get a table like this one

Segment 0 is off in only four rows, so a **decoder** builds it directly: we NOR the outputs for 1, 4, 11 and 13.



a digit display for 0–9 only would make rows 1010–1111
don't-cares